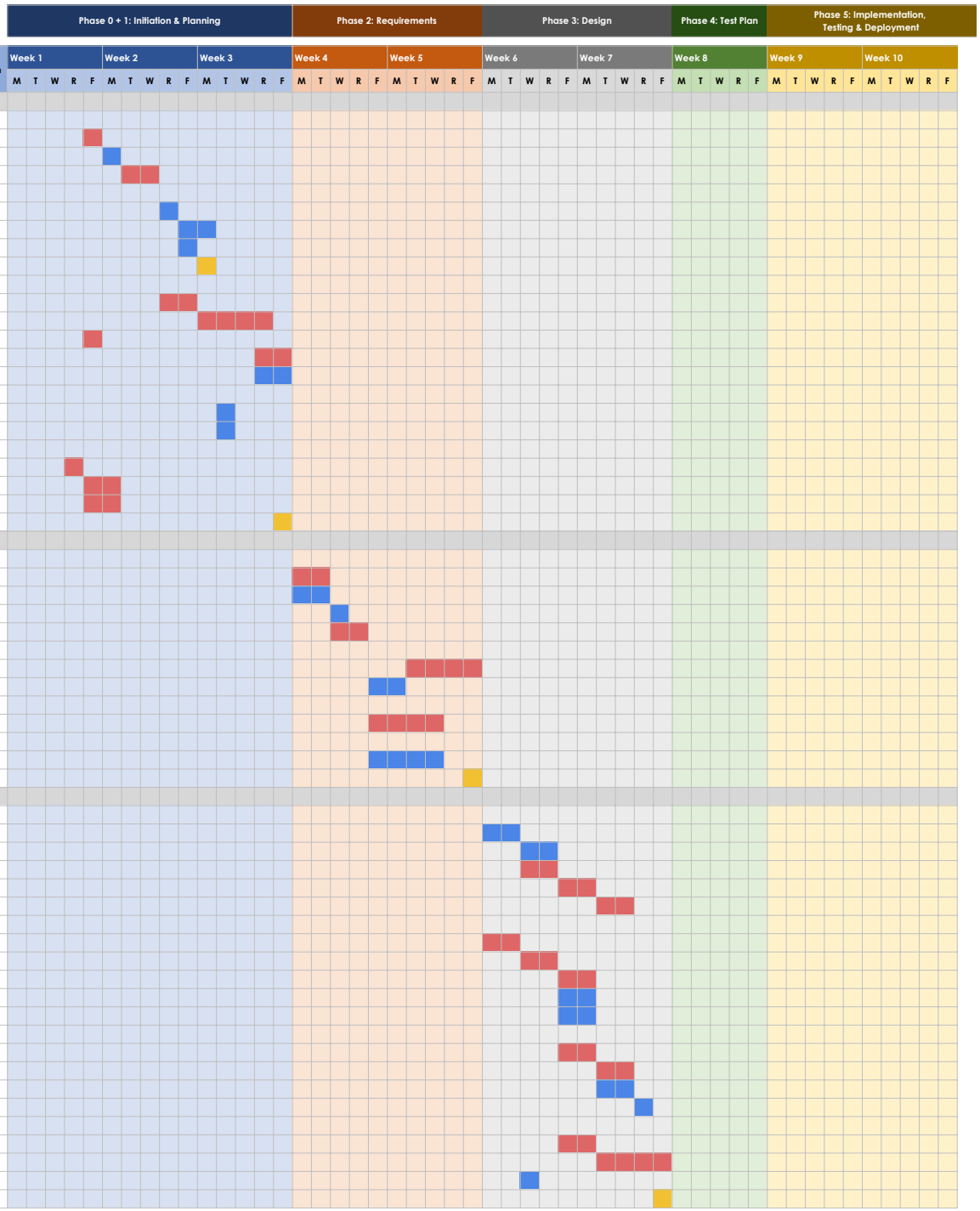
critical path | milestone

WBS Number	Task Title	Task Owner	Start Date	Due Date	Duration
1	**Initiation & planning**		**Oct 7**	**Oct 23**	
1.1	**Vision**				
1.1.1	Define the problem	Lê Đoàn Nhật Huy	Oct 9	Oct 9	1
1.1.2	Identify target users and stakeholders	Lê Đoàn Nhật Huy	Oct 12	Oct 12	1
1.1.3	Define the scope and must-have features	Lê Đoàn Nhật Huy	Oct 13	Oct 14	2
1.2	**Proposed Solution**				
1.2.1	Propose the preliminary application architecture	Trần Hoàng Minh Khang	Oct 15	Oct 15	1
1.2.2	Propose the technology stack	Trần Hoàng Minh Khang	Oct 16	Oct 19	2
1.2.3	Propose the hosting / deployment environment	Trần Hoàng Minh Khang	Oct 16	Oct 16	1
M0	**Milestone: PA0 Project Idea**				
1.3	**Project Plan**				
1.3.1	Define project phases and exit criteria	Huỳnh Trung Ngạn	Oct 15	Oct 16	2
1.3.2	Build a work breakdown structure (WBS)	Huỳnh Trung Ngạn	Oct 19	Oct 22	4
1.3.3	Assign roles and responsibilities	Huỳnh Trung Ngạn	Oct 9	Oct 9	1
1.3.4	Present project schedule	Trương Hiển Long	Oct 22	Oct 23	2
1.3.5	Present work package effort estimation	Trương Hiển Long	Oct 22	Oct 23	2
1.4	**Risk Management**				
1.4.1	Identify technology risks	Trần Hoàng Minh Khang	Oct 20	Oct 20	1
1.4.2	Identify requirements risks	Lê Đoàn Nhật Huy	Oct 20	Oct 20	1
1.5	**Tools & Collaboration Setup**				
1.5.1	Set up communication channels	Huỳnh Trung Ngạn	Oct 8	Oct 8	1
1.5.2	Set up task board	Huỳnh Trung Ngạn	Oct 9	Oct 12	2
1.5.3	Set up collaboration tools	Trương Hiển Long	Oct 9	Oct 12	2
M1	**Milestone: PA1 Project Proposal & Plan Due**				
2	**Requirements**		**Oct 26**	**Nov 6**	
2.1	**Functional, non-functional and domain requirements**				
2.1.1	Define the functional requirements	Lê Đoàn Nhật Huy	Oct 26	Oct 27	2
2.1.2	Define the non-functional requirements	Lê Đoàn Nhật Huy	Oct 26	Oct 27	2
2.1.3	Assign priorities to the requirements	Huỳnh Trung Ngạn	Oct 28	Oct 28	1
2.1.4	Define business rules	Lê Đoàn Nhật Huy	Oct 28	Oct 29	2
2.2	**Use case model**				
2.2.1	List use cases	Lê Đoàn Nhật Huy	Nov 3	Nov 6	4
2.2.2	List actors	Trần Hoàng Minh Khang	Oct 30	Nov 2	2
2.3	**Prototype / mockups**				
2.3.1	Deliver a prototype / mockups	Trần Hoàng Minh Khang	Oct 30	Nov 4	4
2.4	**Traceability**				
2.4.1	Make an RTM (Requirements Traceability Matrix)	Huỳnh Trung Ngạn	Oct 30	Nov 4	4
M2	**Milestone: PA2 Software Requirements Specification Due**				
3	**Design**		**Nov 9**	**Nov 20**	
3.1	**Architectural design**				
3.1.1	Decide on the architectural style	Trần Hoàng Minh Khang	Nov 9	Nov 10	1
3.1.2	Draw the system context diagram (update from PA1)	Trần Hoàng Minh Khang	Nov 11	Nov 12	2
3.1.3	Draw the container diagram	Trần Hoàng Minh Khang	Nov 11	Nov 12	2
3.1.4	Draw the component diagram	Trần Hoàng Minh Khang	Nov 13	Nov 16	2
3.1.5	Draw the deployment Diagram	Trương Hiển Long	Nov 17	Nov 18	2
3.2	**Object-oriented design**				
3.2.1	Draw the conceptual class diagram	Lê Đoàn Nhật Huy	Nov 9	Nov 10	2
3.2.2	Draw the design-level class diagram	Trần Hoàng Minh Khang	Nov 11	Nov 12	2
3.2.3	Draw the sequence diagrams for the main flow	Huỳnh Trung Ngạn	Nov 13	Nov 16	2
3.2.4	Draw the State Machine Diagrams	Huỳnh Trung Ngạn	Nov 13	Nov 16	2
3.2.5	Define Design Principles and Patterns	Trương Hiển Long	Nov 13	Nov 16	2
3.3	**Data design**				
3.3.1	Draw the conceptual ERD of the stored data	Trương Hiển Long	Nov 13	Nov 16	2
3.3.2	Schema Design & Modeling	Trương Hiển Long	Nov 17	Nov 18	2
3.3.3	List Data specifications for DBMS	Trương Hiển Long	Nov 17	Nov 18	2
3.3.4	Identify normal form and the indexes created for frequent queries	Trương Hiển Long	Nov 19	Nov 19	1
3.4	**UI/UX design**				
3.4.1	Draw the screen flow diagram (navigation & routing)	Trần Hoàng Minh Khang	Nov 13	Nov 16	2
3.4.2	Design page views and layout (wireframes)	Trần Hoàng Minh Khang	Nov 17	Nov 20	4
3.4.3	List UX requirements (Accessibility)	Lê Đoàn Nhật Huy	Nov 11	Nov 11	1
M3	**Milestone: PA3 Software Design Due**				



4	**Test planning**		**Nov 23**	**Nov 27**	
4.1	**Test plan**				
4.1.1	List features to be tested or not tested	Lê Đoàn Nhật Huy	Nov 23	Nov 23	1
4.1.2	Map each feature with a test technique	Lê Đoàn Nhật Huy	Nov 24	Nov 24	1
4.1.3	Specify test environment and test data requirements	Trần Hoàng Minh Khang	Nov 23	Nov 24	3
4.2	**Test cases**				
4.2.1	List test cases for each feature/function	Lê Đoàn Nhật Huy	Nov 25	Nov 26	2
4.3	**Traceability**				
4.3.1	Map every Must-Have feature/requirement to >= 1 test case	Huỳnh Trung Ngạn	Nov 27	Nov 27	1
M4	**Milestone: PA4 Test Plan & Test Cases Due**				
5	**Implementation, testing & deployment**		**Nov 30**	**Dec 11**	
5.1	**Pre-implementation**				
5.1.1	Set up and verify the coding environment and toolings	Trương Hiển Long	Nov 30	Nov 30	1
5.1.2	Set up the testing environment	Trương Hiển Long	Dec 1	Dec 1	1
5.1.3	Document coding standards and conventions	Huỳnh Trung Ngạn	Dec 1	Dec 1	1
5.1.4	Configure continuous integration pipeline	Trương Hiển Long	Dec 1	Dec 1	1
5.2	**Frontend Implementation**				
5.2.1	Implement navigation and routing (according to screen flow diagram)	Trần Hoàng Minh Khang	Dec 1	Dec 2	2
5.2.2	Implement page views and layout (according to wireframes)	Trần Hoàng Minh Khang	Dec 3	Dec 4	2
5.2.3	Verify / Implement accessibility (according to UX requirements)	Trần Hoàng Minh Khang	Dec 3	Dec 3	1
5.2.4	Wire CTAs to backend API endpoints	Huỳnh Trung Ngạn	Dec 10	Dec 11	2
5.3	**Database Implementation**				
5.3.1	Install database engine	Trương Hiển Long	Nov 30	Dec 1	2
5.3.2	Configure access controls and credentials	Trương Hiển Long	Dec 2	Dec 2	1
5.3.3	Implement database schema and tables (according to schema design)	Trương Hiển Long	Dec 2	Dec 3	2
5.3.4	Seed test data	Lê Đoàn Nhật Huy	Dec 4	Dec 4	1
5.4	**Backend Implementation**				
5.4.1	Implement authentication and role-based access control (F01)	Trương Hiển Long	Dec 3	Dec 4	2
5.4.2	Implement course management (F02)	Huỳnh Trung Ngạn	Dec 4	Dec 7	2
5.4.3	Implement enrollment and roster management (F03)	Lê Đoàn Nhật Huy	Dec 8	Dec 9	2
5.4.4	Implement course material upload and retrieval (F04)	Trương Hiển Long	Dec 4	Dec 9	4
5.4.5	Implement assignment creation and submission (F05)	Huỳnh Trung Ngạn	Dec 8	Dec 9	2
5.5	**Testing**				
5.5.1	Verify the testing environment and test data	Lê Đoàn Nhật Huy	Nov 30	Nov 30	1
5.5.2	Execute Unit Testing	Trương Hiển Long	Dec 10	Dec 10	1
5.5.3	Document the test results and defects	Lê Đoàn Nhật Huy	Dec 10	Dec 11	1
5.5.4	Set up automated system testing (at least 2 use cases)	Lê Đoàn Nhật Huy	Dec 10	Dec 11	2
5.6	**Deployment**				
5.6.1	Deploy the application to the deployment environment	Trần Hoàng Minh Khang	Dec 10	Dec 11	2
M5	**Milestone: PA5 Implementation, Testing & Deployment Due**				

